Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 02

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. LOAD DATASET
df = pd.read_csv("netflix_titles.csv")

print("=" * 70)
print("NETFLIX DATASET - EXPLORATORY DATA ANALYSIS")
print("=" * 70)

# 2. DISPLAY FIRST RECORDS
print("\nFirst 5 Records:")
print(df.head())

# 3. DATASET DIMENSIONS
print("\nDataset Dimensions:")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
# 4. COLUMN NAMES
print("\nColumn Names:")
print(df.columns.tolist())

# 5. DATA TYPES AND INFORMATION
print("\nDataset Information:")
df.info()

# 6. SUMMARY STATISTICS
print("\nSummary Statistics:")
print(df.describe(include="all"))

# 7. CHECK MISSING VALUES
print("\nMissing Values:")
print(df.isnull().sum())

# 8. VALUE COUNTS
if "type" in df.columns:
    print("\nMovies and TV Shows:")
    print(df["type"].value_counts())

if "rating" in df.columns:
    print("\nContent Ratings:")
    print(df["rating"].value_counts().head(10))

if "release_year" in df.columns:
    print("\nTop Release Years:")
    print(df["release_year"].value_counts().head(10))
```

```
# 9. UNIVARIATE ANALYSIS - DISTRIBUTION OF RELEASE YEAR
if "release_year" in df.columns:
    plt.figure(figsize=(10, 5))

    sns.histplot(
        df["release_year"].dropna(),
        bins=30,
        kde=True
    )

    plt.title("Distribution of Release Years")
    plt.xlabel("Release Year")
    plt.ylabel("Number of Titles")

    plt.show()

# 10. CATEGORICAL VARIABLE ANALYSIS - MOVIES VS TV SHOWS
if "type" in df.columns:
    plt.figure(figsize=(7, 5))

    sns.countplot(
        data=df,
        x="type"
    )

    plt.title("Movies vs TV Shows")
    plt.xlabel("Type")
```

```
    plt.ylabel("Number of Titles")

    plt.show()

# 11. TOP CONTENT RATINGS
if "rating" in df.columns:
    plt.figure(figsize=(10, 5))

    rating_counts = df["rating"].value_counts().head(10)

    sns.barplot(
        x=rating_counts.index,
        y=rating_counts.values
    )

    plt.title("Top Content Ratings")
    plt.xlabel("Rating")
    plt.ylabel("Number of Titles")

    plt.xticks(rotation=45)

    plt.show()

# 12. TOP COUNTRIES
if "country" in df.columns:
    country_counts = (
        df["country"]
        .dropna()
```

```
        .str.split(", ")
        .explode()
        .value_counts()
        .head(10)
    )

    plt.figure(figsize=(10, 5))

    sns.barplot(
        x=country_counts.values,
        y=country_counts.index
    )

    plt.title("Top 10 Countries by Number of Titles")
    plt.xlabel("Number of Titles")
    plt.ylabel("Country")

    plt.show()

# 13. CORRELATION ANALYSIS
numeric_columns = df.select_dtypes(
    include=np.number
).columns

print("\nNumerical Columns:")
print(numeric_columns.tolist())

if len(numeric_columns) >= 2:
```

```
    correlation = df[numeric_columns].corr()

    print("\nCorrelation Matrix:")
    print(correlation)

    plt.figure(figsize=(8, 6))

    sns.heatmap(
        correlation,
        annot=True,
        cmap="coolwarm",
        fmt=".2f"
    )

    plt.title("Correlation Heatmap")

    plt.show()

# 14. BOX PLOT
if "release_year" in df.columns:
    plt.figure(figsize=(8, 5))

    sns.boxplot(
        x=df["release_year"].dropna()
    )

    plt.title("Box Plot of Release Year")
    plt.xlabel("Release Year")
```

```
    plt.show()

# 15. SCATTER PLOT
if "release_year" in df.columns:
    yearly_counts = (
        df["release_year"]
        .value_counts()
        .sort_index()
        .reset_index()
    )

    yearly_counts.columns = [
        "release_year",
        "number_of_titles"
    ]

    plt.figure(figsize=(10, 5))

    sns.scatterplot(
        data=yearly_counts,
        x="release_year",
        y="number_of_titles"
    )

    plt.title("Release Year vs Number of Titles")
    plt.xlabel("Release Year")
    plt.ylabel("Number of Titles")
```

```
    plt.show()

# 16. BIVARIATE ANALYSIS - TYPE VS RATING
if "type" in df.columns and "rating" in df.columns:
    plt.figure(figsize=(12, 6))

    sns.countplot(
        data=df,
        x="rating",
        hue="type",
        order=df["rating"].value_counts().head(10).index
    )

    plt.title("Content Type by Rating")
    plt.xlabel("Rating")
    plt.ylabel("Number of Titles")

    plt.xticks(rotation=45)

    plt.show()

# 17. MULTIVARIATE ANALYSIS
if "type" in df.columns and "release_year" in df.columns:
    yearly_type = (
        df.groupby(
            ["release_year", "type"]
        )
```

```
        .size()
        .reset_index(name="count")
    )

    plt.figure(figsize=(12, 6))

    sns.lineplot(
        data=yearly_type,
        x="release_year",
        y="count",
        hue="type"
    )

    plt.title("Content Type Trends Over Release Years")
    plt.xlabel("Release Year")
    plt.ylabel("Number of Titles")

    plt.show()

# 18. TOP GENRES
if "listed_in" in df.columns:
    genre_counts = (
        df["listed_in"]
        .dropna()
        .str.split(", ")
        .explode()
        .value_counts()
        .head(10)
```

```
    )

    print("\nTop 10 Genres:")
    print(genre_counts)

    plt.figure(figsize=(10, 6))

    sns.barplot(
        x=genre_counts.values,
        y=genre_counts.index
    )

    plt.title("Top 10 Genres")
    plt.xlabel("Number of Titles")
    plt.ylabel("Genre")

    plt.show()

# 19. FINAL OBSERVATIONS
print("\n" + "=" * 70)
print("KEY OBSERVATIONS")
print("=" * 70)

if "type" in df.columns:
    type_counts = df["type"].value_counts()

    print("\n1. Most common content type:")
    print(type_counts.idxmax())
```

```
if "rating" in df.columns:
    print("\n2. Most common content rating:")
    print(df["rating"].mode()[0])

if "release_year" in df.columns:
    print("\n3. Latest release year:")
    print(df["release_year"].max())

    print("\n4. Earliest release year:")
    print(df["release_year"].min())

if "listed_in" in df.columns:
    print("\n5. Most common genre:")
    print(
        df["listed_in"]
        .dropna()
        .str.split(", ")
        .explode()
        .value_counts()
        .idxmax()
    )

print("\nEDA completed successfully!")
```

1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

Answer:
EDA is the process of understanding and analyzing a dataset using statistics and visualizations. It is performed before machine learning to identify patterns, trends, relationships, anomalies, and data-quality issues.

2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Answer:

- **Univariate analysis:** Analysis of one variable, e.g., distribution of release year.
- **Bivariate analysis:** Analysis of two variables, e.g., type and rating.
- **Multivariate analysis:** Analysis of more than two variables, e.g., release year, type, and number of titles.

3. What insights can be obtained from a correlation heatmap?

Answer:
A correlation heatmap shows relationships between numerical variables. It helps identify highly correlated features and shows whether variables have positive or negative relationships.

4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Answer:

- **Histogram:** Shows the distribution and spread of numerical data.
- **Box plot:** Helps identify the spread, median, and outliers.
- **Scatter plot:** Shows the relationship between two variables.

5. How can EDA help identify data quality issues before analysis?

Answer:
EDA helps identify problems such as unusual values, outliers, skewness, missing or inconsistent data, and other potential issues before further analysis.

6. Why is correlation important in predictive analytics? Can correlation imply causation?

Answer:
Correlation is important because it helps identify relationships between variables and useful features for prediction. However, **correlation does not imply causation** because a relationship between two variables does not prove that one causes the other.

7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

Answer:
For categorical variables, **bar charts and count plots** are suitable because they show the frequency of categories. For numerical variables, **histograms and density plots** are suitable because they show data distribution and spread.

8. What business insights can be derived from the Netflix (or Superstore/HR Analytics) dataset through EDA?

Answer:
EDA can identify trends, important patterns, frequent categories, relationships between features, and factors affecting the business objective. These insights can support data-driven business decisions.

9. How does EDA contribute to feature selection and model building?

Answer:
EDA helps identify important relationships between features, highly correlated variables, patterns, and anomalies. This information helps select useful features and prepare better data for model building.

10. What challenges might arise while performing EDA on large-scale real-world datasets?

Answer:
Large datasets may contain a large number of records and features, making analysis and visualization more difficult. Processing large amounts of data can also require more memory and computational resources.

OUTPUT

```
PS D:\PROGRAMS\Career advancement lab\lab sheet 2> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 2\expe2.py"

NETFLIX DATASET - EXPLORATORY DATA ANALYSIS

First 5 Records:
   show_id     type  ...                 listed_in                                description
0        1    Movie  ...        Action, Adventure        An exciting journey across the world.
1        2  TV Show  ...  Drama, International TV Shows  Stories of people living in a modern city.
```

2 3 Movie ... Comedies A family comedy full of unexpected moments.

3 4 Movie ... Dramas, Thrillers A detective uncovers a hidden mystery.

4 5 TV Show ... International TV Shows, Reality TV Chefs compete to create amazing dishes.

[5 rows x 12 columns]

Dataset Dimensions:

Rows: 25

Columns: 12

Column Names:

['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description']

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 25 entries, 0 to 24

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	show_id	25 non-null	int64
1	type	25 non-null	str
2	title	25 non-null	str
3	director	24 non-null	str
4	cast	24 non-null	str
5	country	24 non-null	str
6	date_added	25 non-null	str
7	release_year	25 non-null	int64
8	rating	24 non-null	str
9	duration	25 non-null	str
10	listed_in	25 non-null	str
11	description	25 non-null	str

dtypes: int64(2), str(10)

memory usage: 6.1 KB

Summary Statistics:

	show_id	type	title	...	duration	listed_in	description
count	25.000000	25	25	...	25	25	25
unique	NaN	2	25	...	21	19	25
top	NaN	Movie	The Great Adventure	...	2 Seasons	Documentaries, Nature	An exciting journey across the world.
freq	NaN	17	1	...	3	3	1
mean	13.000000	NaN	NaN	...	NaN	NaN	NaN
std	7.359801	NaN	NaN	...	NaN	NaN	NaN
min	1.000000	NaN	NaN	...	NaN	NaN	NaN
25%	7.000000	NaN	NaN	...	NaN	NaN	NaN
50%	13.000000	NaN	NaN	...	NaN	NaN	NaN
75%	19.000000	NaN	NaN	...	NaN	NaN	NaN
max	25.000000	NaN	NaN	...	NaN	NaN	NaN

[11 rows x 12 columns]

Missing Values:

show_id 0
type 0
title 0
director 1
cast 1
country 1
date_added 0
release_year 0
rating 1
duration 0
listed_in 0
description 0
dtype: int64

Movies and TV Shows:

type

Movie 17

TV Show 8

Name: count, dtype: int64

Content Ratings:

rating

PG 6

PG-13 5

TV-14 3

TV-PG 3

TV-G 2

R 2

G 2

TV-MA 1

Name: count, dtype: int64

Top Release Years:

release_year

2020 4

2022 4

2021 3

2019 3

2023 3

2018 2

2017 2

2024 2

2016 1

2015 1

Name: count, dtype: int64

Numerical Columns:

['show_id', 'release_year']

Correlation Matrix:

	show_id	release_year
show_id	1.000000	-0.016037
release_year	-0.016037	1.000000

Top 10 Genres:

listed_in

Dramas	7
Thrillers	5
Action	4
Documentaries	4
International TV Shows	3
Comedies	3
Nature	3
Reality TV	2
Romantic Movies	2
Docuseries	2

Name: count, dtype: int64

KEY OBSERVATIONS

1. Most common content type:

Movie

2. Most common content rating:

PG

3. Latest release year:

2024

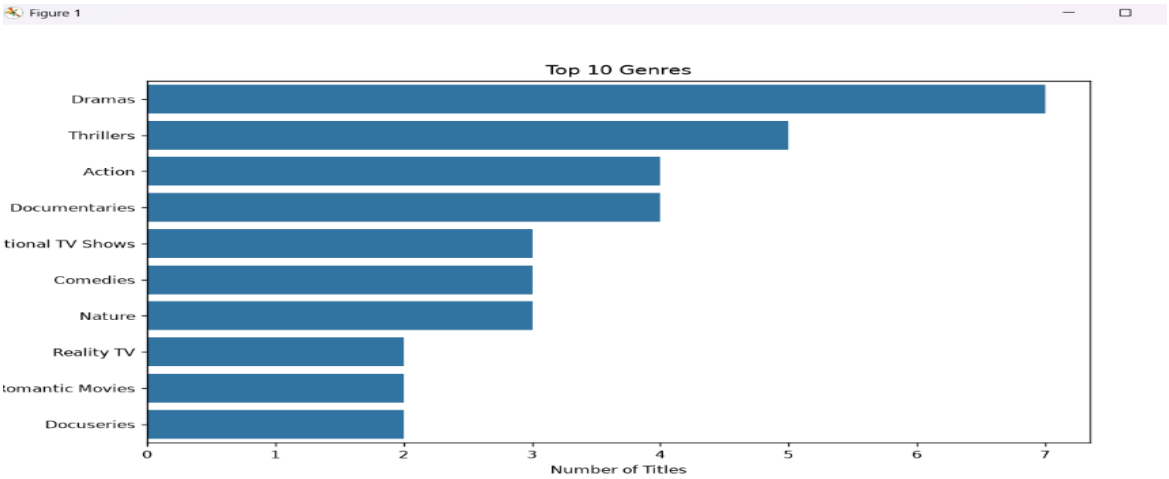
4. Earliest release year:

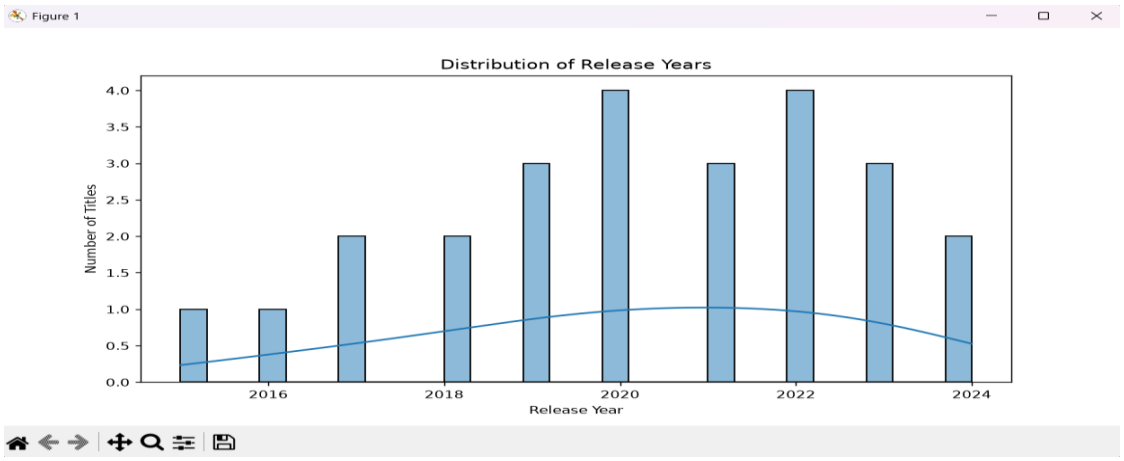
2015

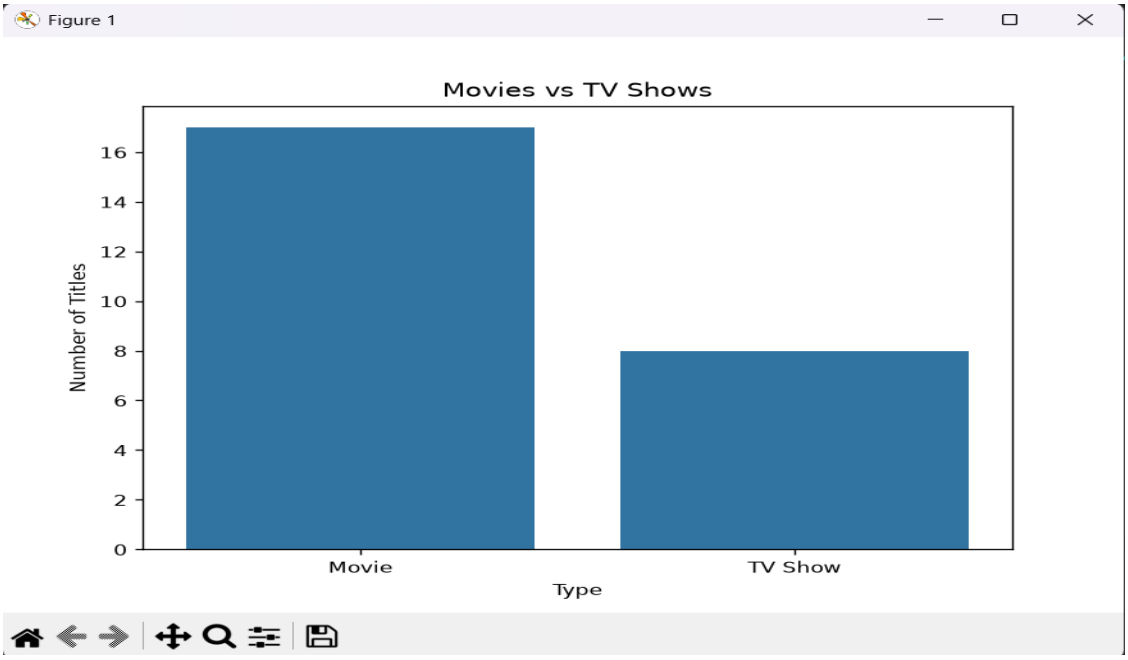
5. Most common genre:

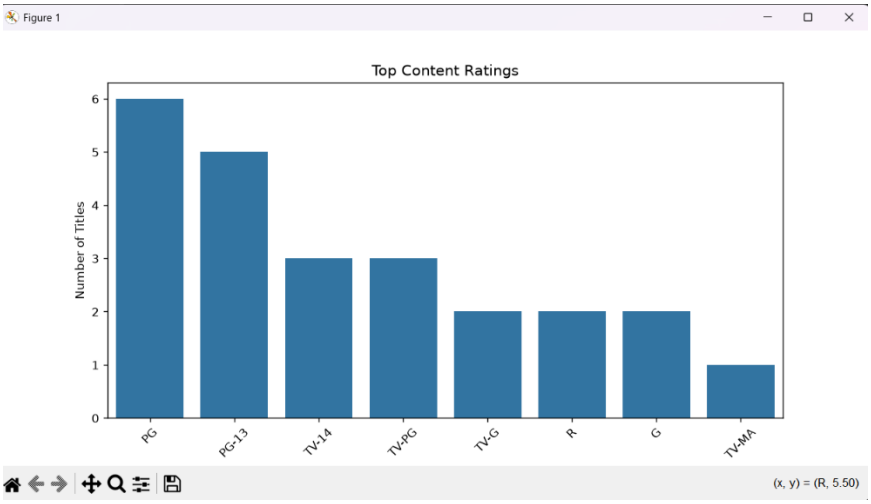
Dramas

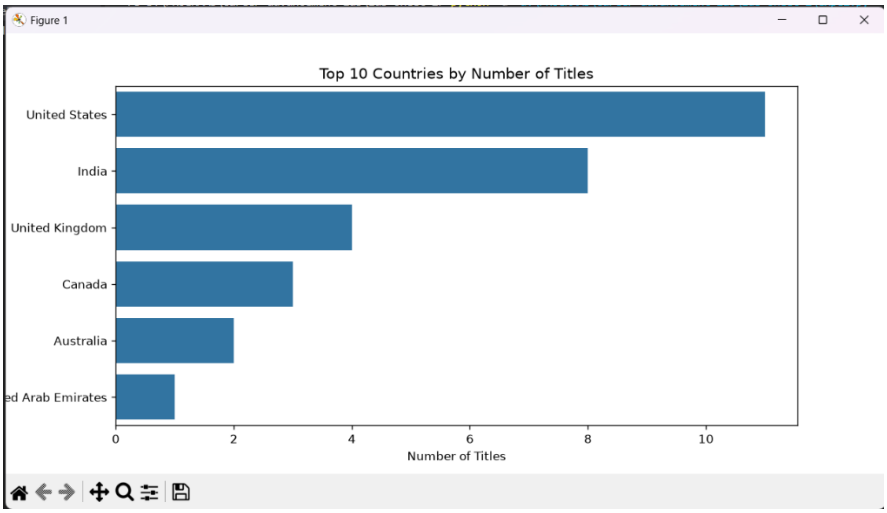
EDA completed successfully!

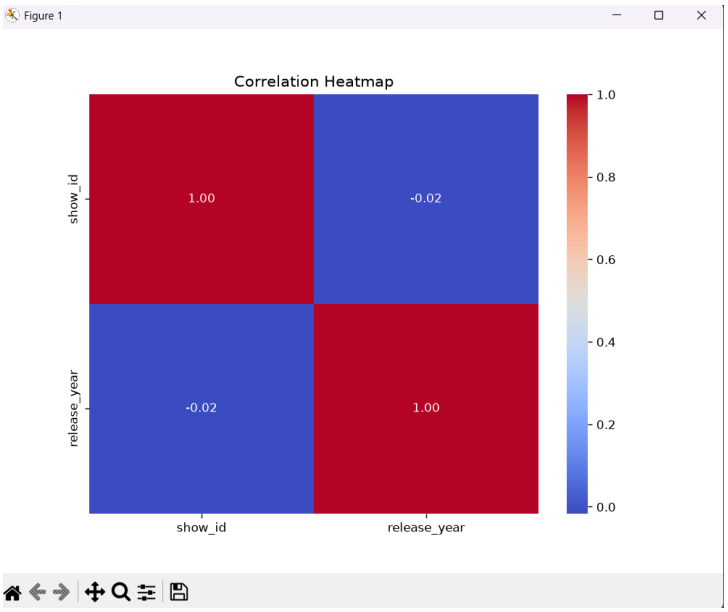

Figure 1
Top 10 Genres
Dramas
Thrillers
Action
Documentaries
tional TV Shows
Comedies
Nature
Reality TV
tomantic Movies
Docuseries
0
1
2
3
4
5
6
7
Number of Titles

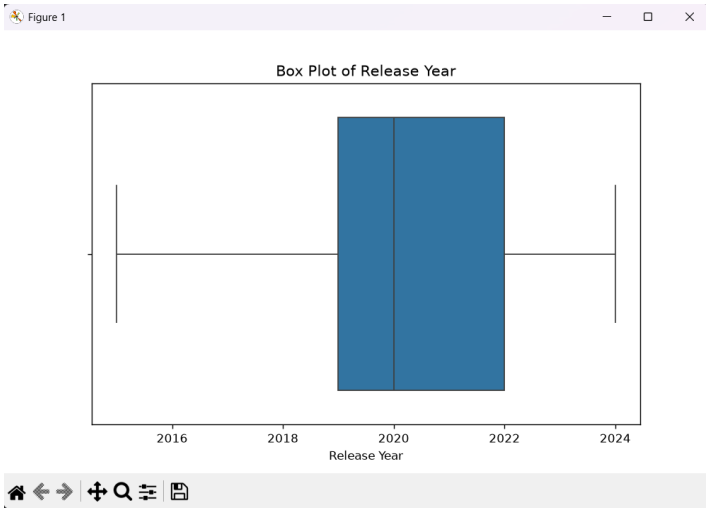

Figure 1
Distribution of Release Years
Number of Titles
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2016
2018
2020
2022
2024
Release Year

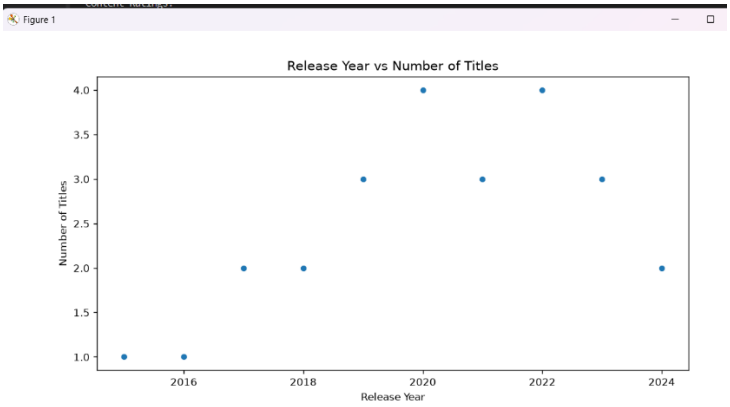

Figure 1
Movies vs TV Shows
Number of Titles
16
14
12
10
8
6
4
2
0
Movie
TV Show
Type

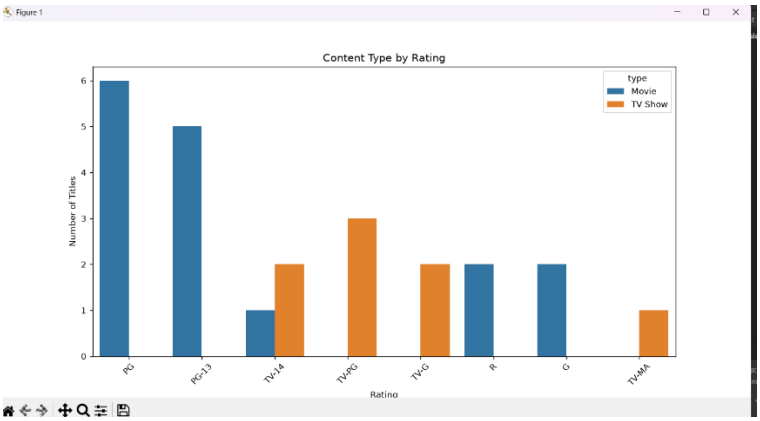

Figure 1
Top Content Ratings
Number of Titles
0
1
2
3
4
5
6
PG
PG-13
TV-14
TV-PG
TV-G
R
G
TV-MA
(x, y) = (R, 5.50)

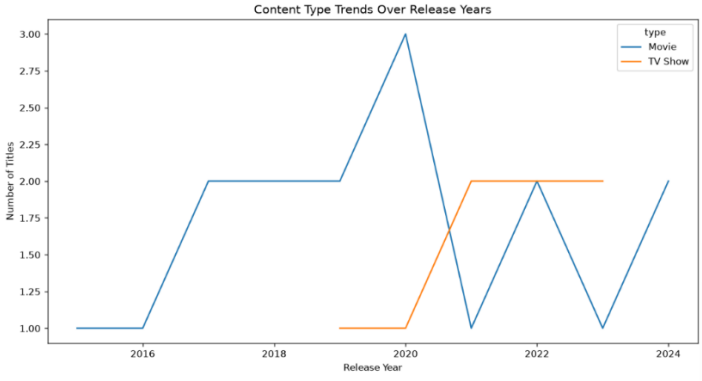

Figure 1
Top 10 Countries by Number of Titles
United States
India
United Kingdom
Canada
Australia
ed Arab Emirates
0
2
4
6
8
10
Number of Titles


Figure 1
Correlation Heatmap
show_id
release_year
1.00
-0.02
-0.02
1.00
show_id
release_year
1.0
0.8
0.6
0.4
0.2
0.0


Figure 1
Box Plot of Release Year
2016
2018
2020
2022
2024
Release Year


Figure 1
Release Year vs Number of Titles
Number of Titles
1.0
1.5
2.0
2.5
3.0
3.5
4.0
2016
2018
2020
2022
2024
Release Year


Figure 1
Content Type by Rating
type
Movie
TV Show
Number of Titles
0
1
2
3
4
5
6
PG
PG-13
TV-14
TV-PG
TV-G
R
G
TV-MA
Rating


Content Type Trends Over Release Years
type
Movie
TV Show
Number of Titles
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
2016
2018
2020
2022
2024
Release Year